

سلام روز بخیر
این متن توسط هوش مصنوعی نوشته شده و سپس اصلاح شده
بوده **AI** تعیین کلاس ها و اعضا و متد ها توسط
ولی من برای این بخش وقتی صرف بررسی و اصلاح متن و یادگیری شیوه پیاده سازی کلاس ها
گذاشتم اما همش رو خودم ننوشتم
امیدوارم که درخور این تلاش نمره ای به این متن بدهید
ممنون

////////////////////////////////////
1

```
+-----+
|          Book          |
+-----+
| - id                   |
| - title                 |
| - author                |
| - subject               |
| - status                | // availability status (available/borrowed)
+-----+
| + setStatus(v)         | // updates the book's status
| + getStatus()          | // returns current status of the book
+-----+
```

```
+-----+
|          Member        |
+-----+
| - memberId              |
| - name                  |
| - phone                 |
| - borrowList            |
+-----+
| + borrowBook(b)        | // borrows a book and adds it to borrowList
| + returnBook(b)        | // returns a borrowed book
+-----+
```

```
+-----+
|        Librarian       |
+-----+
| - employeeId            | // librarian ID
| - name                  |
| - shift                 | // working shift (morning/evening)
+-----+
| + addBook(b)            |
| + removeBook(id)       |
+-----+
```

```
+-----+
|        Borrow          |
+-----+
| - book                  |
| - member                |
| - borrowDate            |
| - returnDate            |
+-----+
| + registerBorrow()      | // logs a borrowing record
| + registerReturn()     | // logs the return of a book
+-----+
```

```

+-----+
|   LibrarySystem   |
+-----+
| - books            |
| - members          |
| - borrows          |
+-----+
| + searchBook()     |
| + lendBook()       | // lends a book to a member
| + returnBook()     |
| + registerMember() | // creates a new member profile
| + showStatus()     | // displays status of all books
+-----+

```

```

////////////////////////////////////
2

```

```

+-----+
|   Product          |
+-----+
| - id               |
| - name             |
| - price            |
| - stock            | // number of available units
+-----+
| + changePrice()    |
| + increaseStock()  | // adds more units to stock
+-----+

```

```

+-----+
|   Customer         |
+-----+
| - id               |
| - name             |
| - email            |
| - cart             | // customer's shopping cart
+-----+
| + placeOrder()     | // creates an order from the cart
| + makePayment()    | // processes customer's payment
+-----+

```

```

+-----+
|   ShoppingCart     |
+-----+
| - items            |
+-----+
| + addItem()        |
| + removeItem()     |
| + calculateTotal() |
+-----+

```

```

+-----+
|   Order            |
+-----+
| - orderId          | // unique order ID
| - customer         |
| - products         |
| - status           | // order status (processing/shipped)
+-----+

```

```

+-----+
| + registerOrder()    |// saves order information
| + changeStatus()    |// updates order status
+-----+

```

```

+-----+
|           Payment    |
+-----+
| - type               |// payment method (card/cash)
| - amount             |// total payable amount
+-----+
| + processPayment()   |// validates and completes payment
+-----+

```

```

+-----+
|           Manager    |
+-----+
| + updateStock()      |
| + changePrice()      |
+-----+

```

```

////////////////////////////////////
3

```

```

+-----+
|           Flight     |
+-----+
| - id                |
| - source             |
| - destination        |
| - date               |
| - capacity           |
+-----+
| + searchFlight()     |// searches for flights matching criteria
| + checkAvailability() |// returns true if seats are available
| + reserveSeat()      |
+-----+

```

```

+-----+
|           Passenger   |
+-----+
| - id                |
| - name              |
| - nationalId         |
+-----+
| + reserveFlight()    |
+-----+

```

```

+-----+
|           Ticket      |
+-----+
| - number             |// unique ticket number
| - flight              |
| - passenger          |
| - status              |
+-----+
| + showInfo()          |// prints ticket details
+-----+

```

```

+-----+
|      Reservation      |
+-----+
| - reservationId      |// unique reservation ID
| - passenger          |// passenger making the reservation
| - flight             |// flight being reserved
| - status             |// status (pending/paid/cancelled)
+-----+
| + payReservation()    |// completes payment and finalizes booking
+-----+

```

```

+-----+
|      Payment          |
+-----+
| - amount             |// payment amount
| - type               |// method (card/wallet)
+-----+
| + processPayment()    |// verifies and processes payment
+-----+

```

```

////////////////////////////////////
4

```

```

+-----+
|      Movie            |
+-----+
| - id                 |
| - title              |
| - duration           |
| - genre              |
+-----+

```

```

+-----+
|      Hall             |
+-----+
| - id                 |// hall number/identifier
| - seatsCount         |
+-----+

```

```

+-----+
|      Show             |
+-----+
| - movie              |
| - hall               |
| - time               |
| - capacity           |
+-----+
| + displayShow()      |// shows movie, hall, and time details
+-----+

```

```

+-----+
|      Ticket           |
+-----+
| - id                 |
| - show               |
| - customerName       |
| - reserved           |// reservation status
+-----+
| + reserveTicket()    |// reserves seat and updates show capacity
+-----+

```

+-----+

+-----+
| CinemaSystem |
+-----+

| - movies |// list of registered movies
| - shows |// list of shows
+-----+
| + searchMovie() |
| + reserveTicket() |// handles ticket reservation
+-----+

////////////////////////////////////
5

+-----+
| Student |
+-----+

| - id |
| - name |
| - courses |
+-----+
| + registerCourse() |// enrolls student in a course
+-----+

+-----+
| Teacher |
+-----+

| - id |
| - name |
| - courses |// courses they teach
+-----+
| + assignGrade() |
+-----+

+-----+
| Course |
+-----+

| - id |
| - title |
| - teacher |
| - students |
+-----+

+-----+
| Grade |
+-----+

| - student |
| - course |
| - value |// grade value (e.g., 18/20)
+-----+

+-----+
| UniversitySystem |
+-----+

| - students |
| - teachers |
| - courses |
+-----+

```
| + registerStudent() |// adds new student to system
| + recordGrade()    |// records student grade
+-----+
```

```
////////////////////////////////////
6
```

```
+-----+
|      Food      |
+-----+
| - id           |
| - name         |
| - price        |
+-----+
```

```
+-----+
|      Restaurant  |
+-----+
| - id            |
| - name          |
| - menu          |// list of Food items
+-----+
```

```
+-----+
|      Customer   |
+-----+
| - id            |
| - name          |
| - address       |// delivery address
+-----+
```

```
+-----+
|      Courier     |
+-----+
| - id            |
| - name          |
| - status        |// free/busy
+-----+
```

```
+-----+
|      Order      |
+-----+
| - customer      |// customer placing order
| - restaurant    |// restaurant fulfilling order
| - foods         |
| - courier       |// assigned courier
| - status        |// order progress (preparing/on the way)
+-----+
```

```
+-----+
|      Manager    |
+-----+
| + addRestaurant() |// registers new restaurant
| + setDiscount()   |// applies discount to menu
+-----+
```

```
////////////////////////////////////
7
```

```

+-----+
|      Station      |
+-----+
| - id              |
| - name            |
| - type            |// type (metro/bus)
+-----+

+-----+
|      Line         |
+-----+
| - id              |
| - stations        |
+-----+

+-----+
|      Vehicle      |
+-----+
| - id              |
| - type            |// type (train/bus)
| - driver          |// assigned driver
+-----+

+-----+
|      Driver       |
+-----+
| - id              |
| - name            |
+-----+

+-----+
|      Card         |
+-----+
| - id              |
| - balance         |// card balance
+-----+
| + charge()        |
| + use()           |// deducts fare on travel
+-----+

////////////////////////////////////
8

```

```

+-----+
| TrafficSystem     |
+-----+
| + controllights() |
+-----+

+-----+
| EnergySystem      |
+-----+
| + monitorUsage()  |
+-----+

+-----+
| SecuritySystem    |
+-----+

```

```

+-----+
| + sendAlert() |
+-----+

```

```

+-----+
|      SmartCity      |
+-----+
| - trafficSystem      |
| - energySystem        |
| - securitySystem      |
+-----+
| + monitorAll()        | // view all systems
+-----+

```

```

////////////////////////////////////
9

```

```

+-----+
|      Member          |
+-----+
| - id                 |
| - name               |
| - membershipType     | // monthly/yearly
+-----+

```

```

+-----+
|      Trainer         |
+-----+
| - id                 |
| - name               |
| - specialty          |
+-----+

```

```

+-----+
|      TrainingClass   |
+-----+
| - id                 |
| - title              |
| - trainer            |
| - members            |
+-----+

```

```

+-----+
|      Payment         |
+-----+
| - id                 |
| - amount              | // amount paid
| - date                |
+-----+

```

```

+-----+
|      GymSystem       |
+-----+
| - members            |
| - classes             | // list of classes
| - payments           |
+-----+

```

```

////////////////////////////////////
10

```

```

+-----+
|      User            |
+-----+

```



```

+-----+
| - id   |
| - name |
| - walletBalance |
+-----+
+-----+
|      Item      |
+-----+
| - id   |
| - title|
| - basePrice|
| - startTime|
| - endTime|
+-----+
+-----+
|      Bid      |
+-----+
| - user   |
| - item   |
| - amount |// bid amount
+-----+
+-----+
|      Auction   |
+-----+
| - item         |// item for auction
| - bids         |// bid list
+-----+
| + placeBid()   |// add bid
| + chooseWinner()// pick top bid
+-----+

```